# CAMPUS TRANSPORT ALLOCATION

## 1. Problem Statement

Develop a Python program to calculate the number of buses required for multiple campus routes based on passenger demand and bus capacity. Identify the number of passengers who cannot be accommodated.

## 2. Objectives

1. Accept passenger demand for each route.
2. Read bus capacity.
3. Calculate the required buses.
4. Identify unserved passengers.
5. Display route-wise transport details.

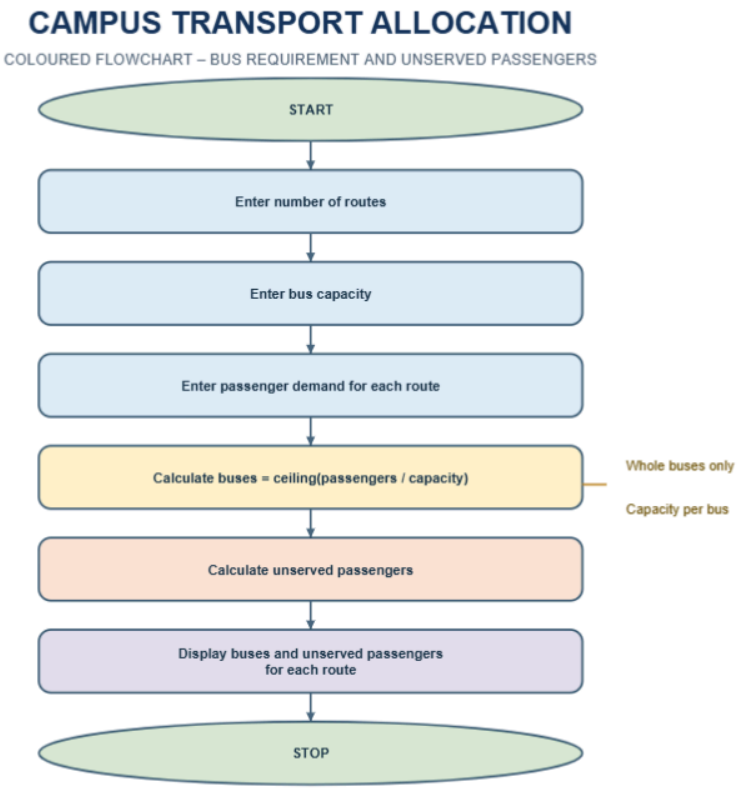
## 3. Assumptions

- Each bus has a capacity of 50 passengers.
- Buses cannot carry more than their capacity.
- Only whole buses are allowed.

## 4. Algorithm

1. Start the program.
2. Enter the number of routes.
3. Enter passenger demand for each route.
4. Enter bus capacity.
5. Calculate required buses using ceiling division.
6. Calculate unserved passengers.
7. Display route-wise results.
8. Stop.

## 5. Flowchart




```
None
                    v
```

## 6. Pseudocode

```
None
START

READ number_of_routes
READ bus_capacity

FOR each route
    READ passengers
    buses = CEILING(passengers / bus_capacity)
    unserved = passengers - (buses * bus_capacity)
```

```
    DISPLAY buses
    DISPLAY unserved
END FOR

STOP
```

## 8. Test Case and Expected Output

Input:

```
None
Enter number of routes: 2
Enter bus capacity: 50

Enter passengers for route: 120
Enter passengers for route: 80
```

Output:

```
None
Route: 1
Required Buses: 3
Unserved Passengers: 0

Route: 2
Required Buses: 2
Unserved Passengers: 0
```

## 9. Conclusion

The program calculates the number of buses required for each campus route based on passenger demand and bus capacity. It also identifies unserved passengers and helps in proper transport allocation.